Neural Network Approximation of American Put Option Prices under Different Volatility Structures

IE 517 Final Project

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Abstract

This project examines how well fully connected neural networks can approximate American put option prices under different volatility structures. Pricing American options is computationally intensive, so we build a three-stage framework to study learnability: (1) constant volatility, (2) a synthetic volatility-smile surface, and (3) an implied-volatility surface calibrated from market data. For each stage, we generate pricing labels using a Partial Differential Equation(PDE) solver and train neural networks on the resulting surfaces. We compare accuracy across volatility regimes and analyze behavior near the early-exercise boundary. Overall, neural networks perform well in smooth, low-volatility settings but become less accurate as volatility curvature and model complexity increase. The results highlight both the promise and the limitations of neural networks as fast surrogate pricers for American options.

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1 Introduction

Pricing American-style options remains a challenging problem in quantitative finance because the early-exercise feature introduces a free-boundary condition and strong nonlinearities in the value function. Classical numerical methods—such as finite-difference schemes, binomial

trees, or Longstaff–Schwartz Monte Carlo—can deliver accurate prices but are often too slow for real-time decision making or large-scale simulations.

Recent work suggests that deep learning models may serve as efficient surrogate pricers by learning the underlying pricing function from data generated by traditional solvers. Once trained, a neural network can evaluate new option prices almost instantly, making it attractive for tasks such as scenario generation, calibration loops, and high-frequency risk computations. However, the effectiveness of these models depends heavily on the structure and quality of the volatility input, which strongly influences the shape and complexity of the pricing surface.

To study this systematically, we design a three-stage framework that increases in volatility complexity: (1) a constant-volatility baseline, (2) a synthetic volatility-smile surface, and (3) a market-calibrated implied-volatility surface. Each stage allows us to evaluate how volatility assumptions affect learnability, accuracy, and generalization. By comparing neural-network performance across these settings, we aim to understand when deep learning can reliably approximate American option prices and where model limitations begin to appear.

This project studies:

- How well neural networks can learn the American put pricing surface,
- how different volatility structures affect the difficulty of learning, and
- how robust the model remains across different volatility regimes.

2 Stage 1: Scalar Volatility

This stage establishes a baseline for neural-network surrogate pricing of American put options under scalar volatility assumptions. All pricing results in this stage are generated under controlled numerical settings and serve as a reference point for the more complex volatility surfaces introduced later.

2.1 Stage 1 Setup

2.1.1 Market Environment

We consider a single underlying asset with spot price $S_0 = 100$. The risk-free interest rate and dividend yield are assumed to be flat across maturities, with $r = 3\%$ and $q = 0$. These assumptions are fixed throughout Stage 1 to isolate model-driven effects rather than market-structure variations. All term structures are implemented using flat curves in `QuantLib`.

2.1.2 Input Domain and Sampling

The pricing domain is defined over strike K, maturity T, and volatility level σ. Table 1 summarizes the discretization scheme. The Cartesian product of these grids yields 1050 pricing points, which form the supervised-learning dataset.

Table 1: Input domain for Stage 1 scalar-volatility experiments.

Variable	**Range / Values**	**Grid Size**
Strike K	$[60, 140]$ (evenly spaced)	21
Maturity T (years)	$[0.25, 2.0]$	10
Volatility σ	$\{0.15, 0.20, 0.25, 0.30, 0.40\}$	5
Total points	$21 \times 10 \times 5 = 1050$	

2.1.3 Benchmark PDE Pricers

For each (K, T, σ) combination, American put prices are computed under two models:

- **Black–Scholes**, using `FdBlackScholesVanillaEngine`;
- **Heston**, using `FdHestonVanillaEngine`.

To ensure comparability between the two models, the Heston parameters are chosen such that

$$v_0 = \theta = \sigma^2,$$

so both models share the same effective volatility level. This design isolates the impact of stochastic-volatility dynamics rather than differences in average volatility.

Each input point therefore has two aligned benchmark labels, $V_{\mathrm{BS}}^{\mathrm{Am}}$ and $V_{\mathrm{Heston}}^{\mathrm{Am}}$, which are used as ground truth for surrogate training and evaluation.

2.2 Learning Objective and Model Design

2.2.1 Learning Targets

Two neural-network surrogate pricers are trained:

- $\mathrm{NN}_{\mathrm{BS}}$: $(K, T, \sigma) \mapsto V_{\mathrm{BS}}^{\mathrm{Am}}$;
- $\mathrm{NN}_{\mathrm{Heston}}$: $(K, T, \sigma) \mapsto V_{\mathrm{Heston}}^{\mathrm{Am}}$.

The objective is to compare approximation difficulty and generalization behavior between constant-volatility and stochastic-volatility pricing surfaces under identical network architectures and data splits.

2.2.2 Neural Network Architecture

Both surrogates share the same fully connected architecture, summarized in Table 2. This controlled design ensures that performance differences can be attributed to model complexity rather than architectural choices.

Table 2: Neural network architecture used in Stage 1.

Component	**Specification**
Input dimension	3
Hidden layers	3
Hidden width	64
Activation	ReLU
Output dimension	1

2.2.3 Training Configuration

Training settings are identical for both networks and are reported in Table 3. All experiments are conducted on CPU.

Table 3: Training configuration for Stage 1.

Setting	**Value**
Loss function	Mean squared error
Optimizer	Adam
Learning rate	10^{-3}
Batch size	64
Epochs	500

Both models converge smoothly, typically reaching stable low training loss within 150–200 epochs.

2.3 Overall Approximation Accuracy

Table 4 reports test-set performance on aligned evaluation points.

Table 4: Overall test performance in Stage 1.

Model	**MAE**	**RMSE**
$\mathrm{NN}_{\mathrm{BS}}$	0.0837	0.1120
$\mathrm{NN}_{\mathrm{Heston}}$	0.1453	0.1971

The larger error of the Heston surrogate reflects the increased nonlinearity induced by stochastic volatility dynamics.

2.4 Experiment I: Volatility-Regime Learning Difficulty

To better understand how approximation difficulty varies across volatility regimes, we group test-set samples into three volatility buckets based on the input volatility level σ. This experiment examines whether surrogate errors are uniformly distributed across regimes or concentrated in specific volatility ranges.

The test set spans volatility levels in the range $\sigma \in [0.15, 0.40]$. Table 5 reports detailed error statistics and sample counts for each volatility bucket, followed by a visual comparison in Figure 1.

Table 5: Test-set error by volatility regime.

Volatility Bucket	**Model**	**# Points**	**MAE**	**RMSE**
Low $[0.10, 0.20]$	Black–Scholes	88	0.0794	0.1085
	Heston	88	0.1223	0.1668
Medium $(0.20, 0.35]$	Black–Scholes	80	0.0834	0.1092
	Heston	80	0.1350	0.1912
High $(0.35, 0.50]$	Black–Scholes	42	0.0933	0.1238
	Heston	42	0.2134	0.2575

Table 5 reveals two clear and systematic patterns. For the Black–Scholes surrogate, approximation errors are relatively stable across volatility regimes, with only a mild increase from low to high volatility. This behavior is consistent with the smooth structure of the Black–Scholes American pricing surface in (K, T, σ), even at elevated volatility levels.

In contrast, the Heston surrogate exhibits a pronounced volatility-dependent deterioration in accuracy. From the low- to the high-volatility bucket, both MAE and RMSE increase by approximately 50–70%, and in the high-volatility regime the Heston errors are nearly twice those of the Black–Scholes surrogate at the same volatility level. This reflects the additional nonlinearity and curvature introduced by stochastic volatility dynamics, which substantially increase the complexity of the American option pricing surface and make it harder for a simple feed-forward network to approximate.

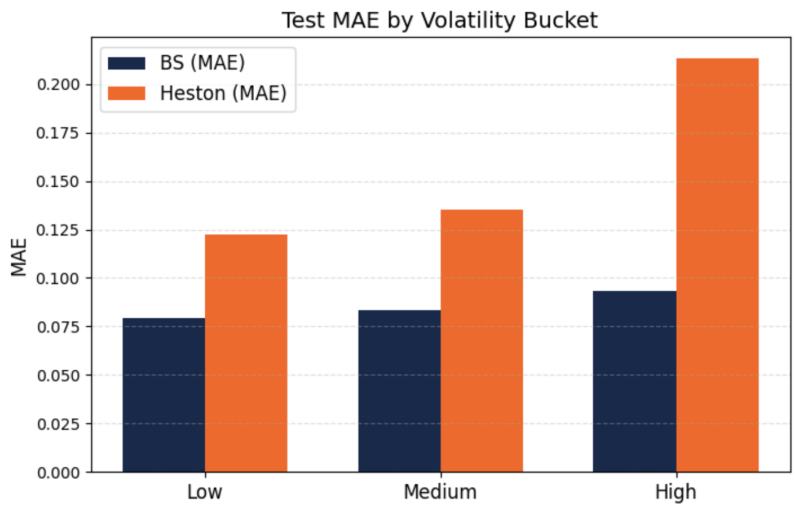

Figure 1: Test-set MAE by volatility bucket for Black–Scholes and Heston surrogate models.

Figure 1 visually reinforces these findings, clearly illustrating the stability of the Black–Scholes errors across regimes and the sharp increase in Heston error in the high-volatility bucket.

2.5 Experiment II: Price Surfaces and Cross-Model Differences

This experiment analyzes how Black–Scholes and Heston American put prices differ when both models are evaluated on the same (K, T, σ) domain. Rather than focusing on absolute pricing accuracy, the goal is to understand the structural pricing gap between the two models and to assess whether neural networks can reproduce this cross-model difference.

We define the price difference

$$\Delta V(K, T, \sigma) = V_{\text{Heston}}(K, T, \sigma) - V_{\text{BS}}(K, T, \sigma).$$

Because both models share the same effective volatility level (with $v_0 = \theta = \sigma^2$ in the Heston specification), the sign and magnitude of ΔV reflect differences arising from stochastic variance dynamics rather than from volatility level mismatches.

2.5.1 Volatility-Bucket Statistics of ΔV

We first summarize the distribution of ΔV and the neural approximation error across volatility regimes. Table 6 reports statistics on the test set, grouped by volatility bucket.

Table 6: Cross-model price difference statistics by volatility regime.

Volatility Bucket	**# Points**	**Mean ΔV**	**Min**	**Max**	**MAE**	**RMSE**
Low $[0.10, 0.20]$	88	-0.0446	-0.7735	0.5802	0.1803	0.2402
Medium $(0.20, 0.35]$	80	-0.0371	-0.9726	0.5405	0.1826	0.2505
High $(0.35, 0.50]$	42	-0.3241	-1.0828	0.4439	0.2803	0.3531
All	210	-0.0976	-1.0828	0.5802	0.2012	0.2702

Several observations follow directly from Table 6. Across all regimes, the average price difference ΔV is negative, indicating that Heston generally produces lower American put prices than Black–Scholes under matched volatility. This effect is relatively mild at low and medium volatility but becomes substantially stronger in the high-volatility regime, where the mean ΔV is nearly an order of magnitude larger in absolute value. In addition, neural approximation errors increase with volatility, suggesting that the structural model gap itself becomes more complex at higher volatility levels.

2.5.2 Cross-Model Price Difference Surfaces

To visualize the structure of the model gap, we plot heatmaps of $\Delta V(K, T)$ at fixed volatility levels $\sigma = 0.20, 0.30$, and 0.40. For each volatility level, we compare the true price difference surface with its neural-network approximation $\widehat{\Delta V}$.

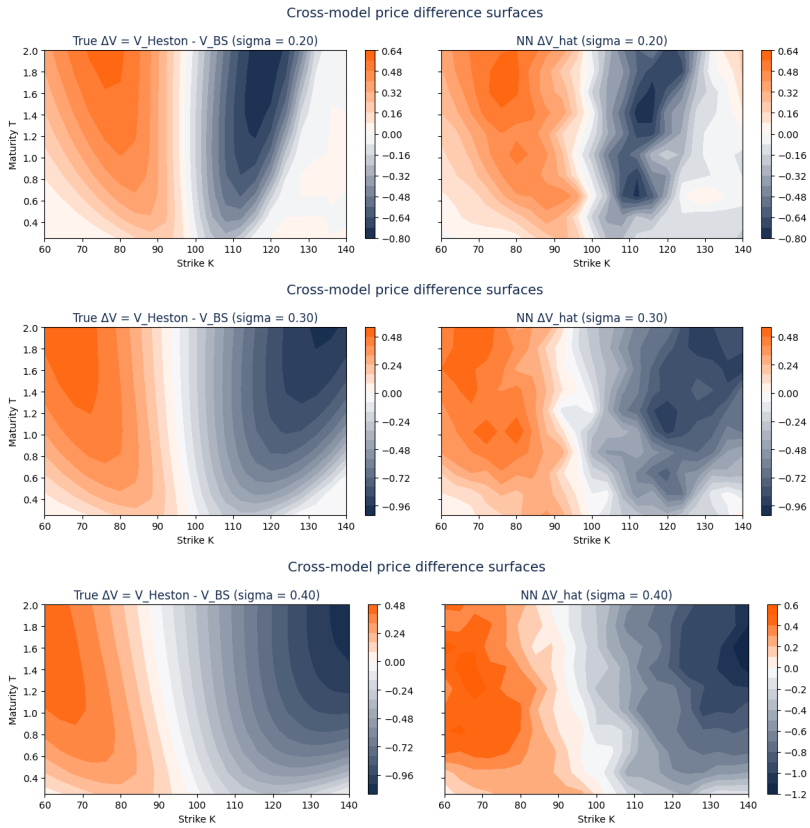

Figure 2: Cross-model price difference surfaces $\Delta V = V_{\text{Heston}} - V_{\text{BS}}$ (left) and neural approximation $\widehat{\Delta V}$ (right) at fixed volatility levels.

The heatmaps reveal that the cross-model difference is highly structured rather than a constant offset. At lower volatility, ΔV varies smoothly across strike and maturity, while at higher volatility the surface exhibits stronger curvature and sharper transitions across moneyness. The neural networks successfully capture the dominant shape and sign structure of the gap, although local distortions become more pronounced as volatility increases.

2.5.3 Maturity Slices at Fixed Strike

To further examine the dependence of the pricing gap on maturity and moneyness, we plot one-dimensional slices of ΔV and $\widehat{\Delta V}$ as functions of maturity T, at an at-the-money strike ($K = 100$) and a deep in-the-money strike ($K = 120$).

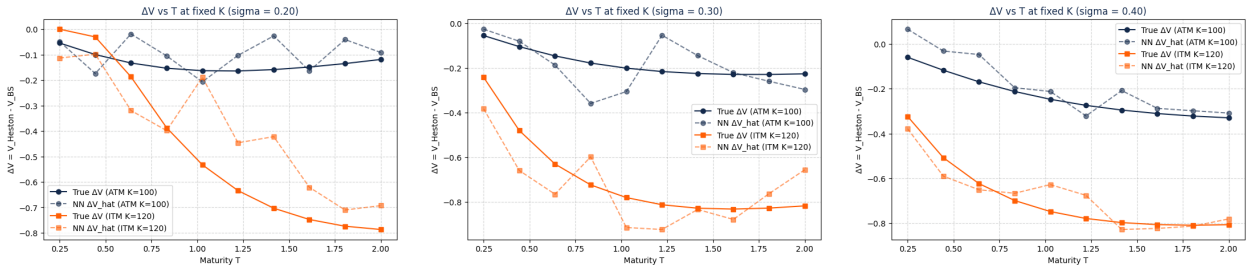

Figure 3: Maturity slices of the cross-model price difference at fixed strikes, comparing true ΔV and neural approximation $\widehat{\Delta V}$ for $\sigma = 0.20, 0.30$, and 0.40.

These slices highlight several systematic effects. First, the magnitude of ΔV generally increases with maturity, particularly for deep in-the-money options, indicating that stochastic volatility effects accumulate over longer horizons. Second, the pricing gap is substantially larger for in-the-money options than for at-the-money options, reflecting the interaction between early-exercise incentives and variance dynamics. The neural approximation tracks these trends well but exhibits increased deviation at long maturities and high volatility, consistent with the surface-level results.

2.5.4 Experiment II Discussion

Overall, Experiment II demonstrates that the difference between Black–Scholes and Heston American put prices is a structured, regime-dependent surface rather than a simple bias. The gap is generally negative and becomes significantly more pronounced in high-volatility, long-maturity, and in-the-money regions. Neural networks are capable of learning the main geometry of this cross-model difference, but approximation errors increase precisely in the regimes where stochastic volatility introduces the strongest nonlinear effects.

These findings complement the results of Experiment I by showing that not only is Heston pricing harder to approximate in absolute terms, but the *relative* difference between Heston and Black–Scholes itself becomes increasingly complex as volatility rises.

2.6 Experiment III: Runtime and Accuracy Benefits of Neural Networks

In the final experiment, we quantify the computational gains from using neural networks as surrogate pricers and examine how these gains depend on model complexity. We randomly sample $N = 2000$ points (K, T, σ) from the same pricing domain as before and evaluate four pricing engines: (i) Black–Scholes finite-difference pricer (BS PDE), (ii) Heston

finite-difference pricer (Heston PDE), (iii) neural surrogate $\mathrm{NN}_{\mathrm{BS}}$, and (iv) neural surrogate $\mathrm{NN}_{\mathrm{Heston}}$.

For each method, we measure total runtime and average time per pricing query. In addition, we evaluate surrogate accuracy by comparing $\mathrm{NN}_{\mathrm{BS}}$ and $\mathrm{NN}_{\mathrm{Heston}}$ against their corresponding PDE benchmarks on the same random sample, reporting MAE and RMSE. The goal is to demonstrate that while Heston PDE pricing is substantially slower than BS PDE, neural inference time is nearly identical across surrogates and orders of magnitude faster than the underlying PDE solvers.

2.6.1 Runtime Summary

Table 7 reports total runtime and time-per-point for each pricing engine on the random sample.

Table 7: Runtime comparison on a random sample of $N = 2000$ pricing points.

Method	**Total Time (s)**	**Time per Point (s)**
BS PDE	8.1392	0.004070
Heston PDE	383.5455	0.191773
$\mathrm{NN}_{\mathrm{BS}}$	0.003608	1.80×10^{-6}
$\mathrm{NN}_{\mathrm{Heston}}$	0.001414	7.10×10^{-7}

Table 7 shows that Heston PDE pricing is orders of magnitude slower than Black–Scholes PDE pricing on the same input domain. In contrast, both neural surrogates operate at microsecond scale, and their inference times are of the same order regardless of whether the target surface comes from Black–Scholes or Heston. This highlights that the computational advantage of surrogate pricing is especially pronounced for complex stochastic-volatility models.

2.6.2 Runtime Visualizations

To make the scale differences explicit, Figure 4 summarizes runtime comparisons across PDE solvers and neural surrogates.

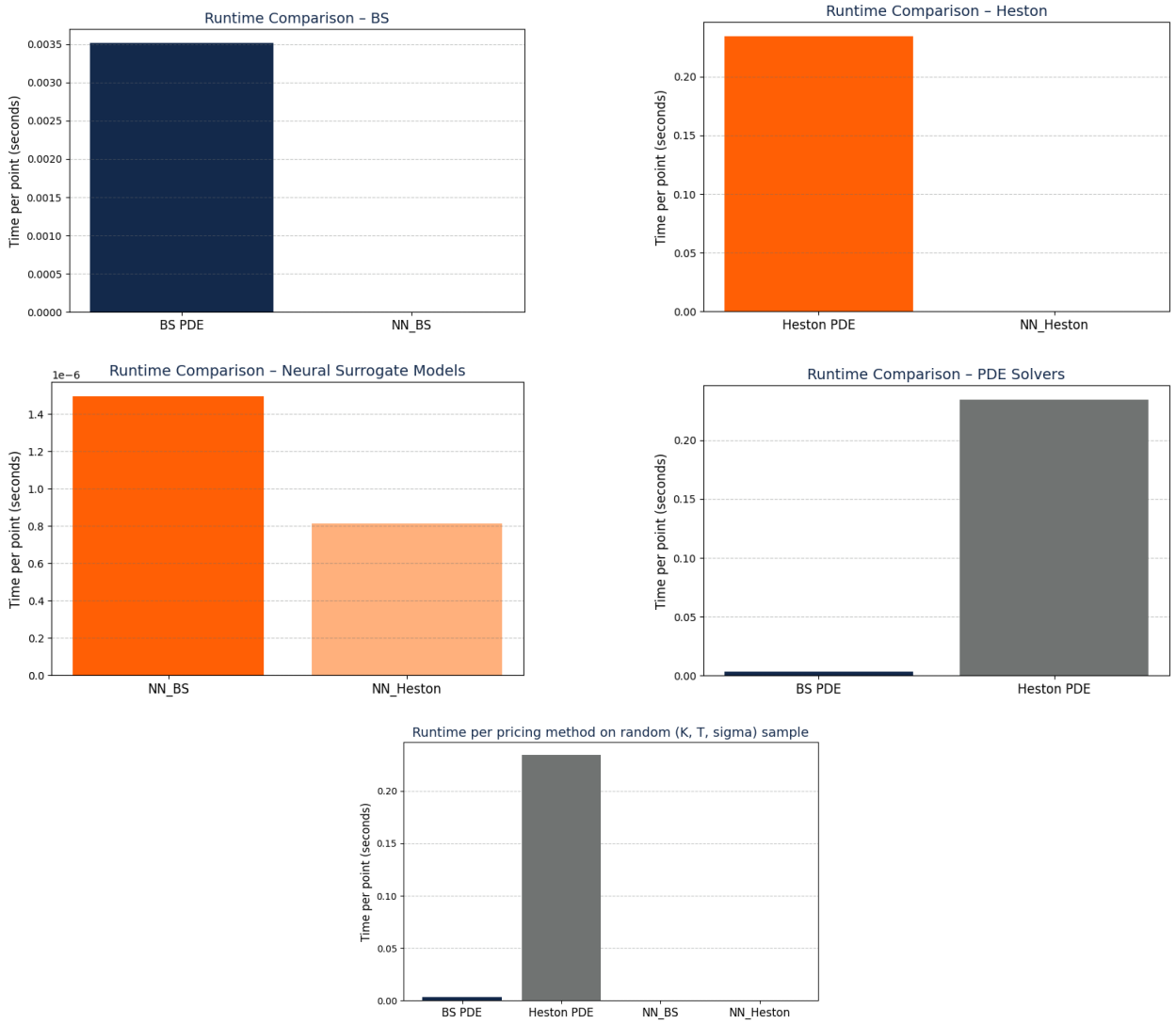

Figure 4: Runtime comparisons across PDE solvers and neural surrogates. Top: BS PDE vs. $\mathrm{NN}_{\mathrm{BS}}$ and Heston PDE vs. $\mathrm{NN}_{\mathrm{Heston}}$. Middle: neural surrogates and PDE solvers compared within their respective groups. Bottom: combined comparison across all four methods.

Figure 4 visually reinforces the main conclusion: Heston PDE is the dominant computational bottleneck, whereas neural inference remains negligible on the same scale. In particular, PDE runtimes span milliseconds to tenths of seconds per query, while neural surrogates operate at microsecond scale, creating a separation of several orders of magnitude.

2.6.3 Accuracy of Neural Surrogates

Table 8 reports pricing accuracy of neural surrogates relative to their PDE benchmarks on the same random sample.

Table 8: Surrogate accuracy relative to PDE benchmarks on the random sample.

Surrogate	**MAE**	**RMSE**
$\mathrm{NN}_{\mathrm{BS}}$ (vs. BS PDE)	0.080701	0.109899
$\mathrm{NN}_{\mathrm{Heston}}$ (vs. Heston PDE)	0.141181	0.189133

These results demonstrate a clear speed–accuracy trade-off: neural surrogates reduce per-query runtime by orders of magnitude while maintaining pricing errors on the order of 10^{-1}. Consistent with Experiments I–II, the Heston surrogate exhibits larger errors due to the increased complexity of the stochastic-volatility pricing surface.

2.6.4 Experiment III Discussion

Overall, Experiment III confirms that neural surrogate pricing provides dramatic computational acceleration, particularly for stochastic-volatility models where traditional PDE solvers are expensive. While Heston PDE requires roughly two orders of magnitude more time per query than BS PDE, the inference time of $\mathrm{NN}_{\mathrm{BS}}$ and $\mathrm{NN}_{\mathrm{Heston}}$ remains nearly identical and negligible on the PDE scale. Combined with the accuracy results, this supports the use of surrogate models as practical replacements for repeated pricing queries in calibration, risk, and real-time evaluation settings.

2.7 Stage I Summary: Scalar-Volatility Baseline

Stage I establishes a scalar-volatility baseline on a controlled (K, T, σ) domain. Here, both Black–Scholes and Heston American put prices are evaluated under matched effective volatility (via $v_0 = \theta = \sigma^2$ in the Heston specification), so that any differences reflect stochastic-variance dynamics rather than volatility-level discrepancies. This baseline is essential because subsequent stages replace the scalar σ input with non-scalar volatility structures (e.g.,

implied-volatility smiles and surfaces), and Stage I provides the reference behavior against which those extensions are interpreted.

Experiment I showed that under scalar volatility, the Black–Scholes pricing surface remains comparatively smooth across volatility regimes, while the Heston surface becomes substantially harder to approximate in high-σ regions. This volatility-wise difficulty profile quantifies how stochastic volatility increases surface nonlinearity even when the volatility level is held fixed in expectation.

Experiment II analyzed the structural cross-model gap $\Delta V = V_{\text{Heston}} - V_{\text{BS}}$. We found that ΔV is not a constant bias but a regime-dependent surface that varies systematically with moneyness, maturity, and volatility level. In particular, the gap becomes more negative and more structured in high-volatility and in-the-money regions, highlighting where stochastic-variance dynamics and early-exercise effects interact most strongly under scalar-volatility matching.

Finally, Experiment III quantified the computational implications of the scalar-volatility baseline. While Heston PDE pricing is orders of magnitude slower than Black–Scholes PDE pricing, neural surrogates provide microsecond-level inference for both models. Crucially, the surrogate inference cost is largely model-agnostic, so the speedup is most dramatic precisely when the underlying PDE baseline is most expensive (e.g., Heston), motivating the use of surrogates for downstream tasks such as repeated pricing queries, calibration loops, and real-time evaluation.

Overall, Stage I provides a clean scalar-volatility reference for both model behavior and computational cost, which we use as the baseline when moving to non-scalar volatility representations in later stages.

3 Stage 2: Synthetic Volatility-Smile Surface

3.1 Motivation

In Stage 1, volatility was treated as an exogenous scalar input, and American put prices were computed under fixed volatility regimes. While this setup is useful for controlled experiments, it does not reflect the empirical fact that volatility varies systematically across strikes and maturities.

To bridge this gap, Stage 2 introduces a deterministic volatility surface and examines how such structure propagates through the American option pricing PDE. Importantly, we do not alter the pricing model or numerical method; instead, we isolate the effect of volatility specification by holding the PDE solver fixed.

3.2 Deterministic Volatility Surface Specification

We define the following parametric volatility surface:

$$\sigma(K,T) = 0.20 + 0.10 \left| \log\left(\frac{K}{F_0}\right) \right| + 0.05\,T, \tag{1}$$

where F_0 denotes the forward level of the underlying.

This specification embeds two stylized empirical features. First, the absolute log-moneyness term generates a symmetric volatility smile across strikes. Second, the linear maturity term produces a mildly upward-sloping term structure, reflecting increasing uncertainty over longer maturities.

The volatility surface is fully deterministic and is not calibrated to market data. Its purpose is to serve as a controlled benchmark that introduces realistic volatility structure without additional estimation noise.

3.3 American Put Pricing under Local Volatility

Using the constructed volatility surface, we compute American put prices by solving the Black–Scholes PDE with pointwise local volatility:

$$V_{\text{locvol}}(K,T) = V_{\text{BS}}^{\text{Am}}(K,T;\,\sigma(K,T))\,. \tag{2}$$

At each (K,T) grid point, the local volatility value $\sigma(K,T)$ is evaluated and supplied to the finite-difference solver. The spatial and temporal grids, boundary conditions, and early-exercise treatment are identical to those used in Stage 1. As a result, any differences between the Stage 1 and Stage 2 price surfaces can be attributed solely to the introduction of the volatility surface.

3.4 Comparison with Constant-Volatility and Heston Surfaces

We compare the American put price surface under deterministic local volatility with two benchmark models: (i) the constant-volatility Black–Scholes American surface from Stage 1, (ii) the Heston American surface computed via a two-dimensional PDE solver.

Relative to the constant-volatility case, the local-volatility surface exhibits pronounced strike-dependent curvature, particularly at short maturities, reflecting the direct impact of the embedded volatility smile. However, unlike the Heston surface, the local-volatility surface remains symmetric by construction and does not generate the left skew induced by stochastic variance dynamics.

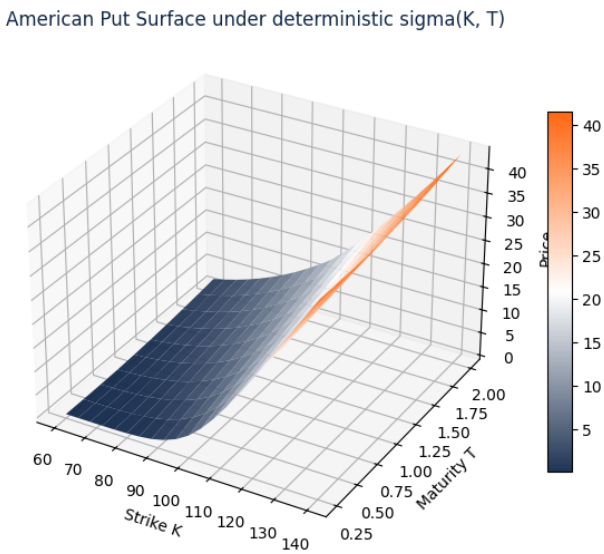

Figure 5: American put price surface under local volatility

This comparison highlights that deterministic volatility structure alone canvsubstantially reshape American option prices, while stochastic volatility is required to produce asymmetric pricing patterns observed in practice.

3.5 Computational Cost

We measure the average runtime per pricing point for each method under identical grid and numerical settings. The results are summarized in Table 9.

Table 9: Average computational cost per pricing point.

Method	PDE Dimension	Volatility Specification	Time per Point (s)	Relative Cost
BS PDE	1D (S)	Constant / Local	0.0057	1× (baseline)
Heston PDE	2D (S, v)	Stochastic	0.2390	≈ 42×

The Black–Scholes PDE solver requires approximately 0.0057 seconds per pricing point, while the Heston PDE solver requires approximately 0.2390 seconds, corresponding to a computational cost roughly 42 times higher. This difference is primarily driven by the additional variance dimension in the Heston PDE formulation.

Importantly, introducing a deterministic volatility surface within the Black–Scholes framework does not increase the dimensionality of the underlying PDE and therefore leaves its computational cost essentially unchanged. This result highlights the computational advantage of deterministic local volatility models relative to stochastic volatility specifications.

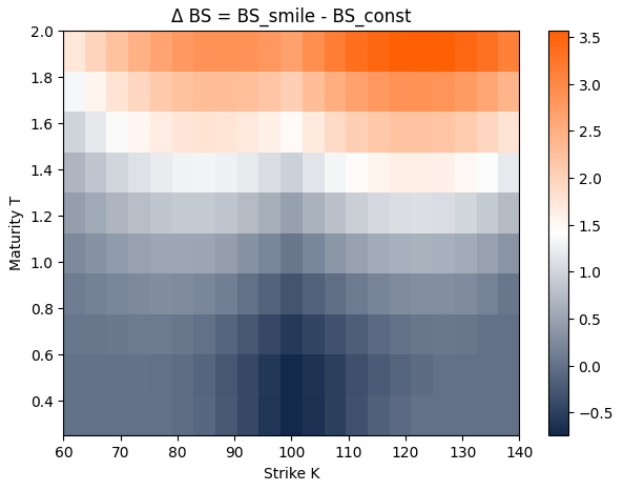

Figure 6: Local-Volatility vs Constant-Volatility Black–Scholes American Put Prices

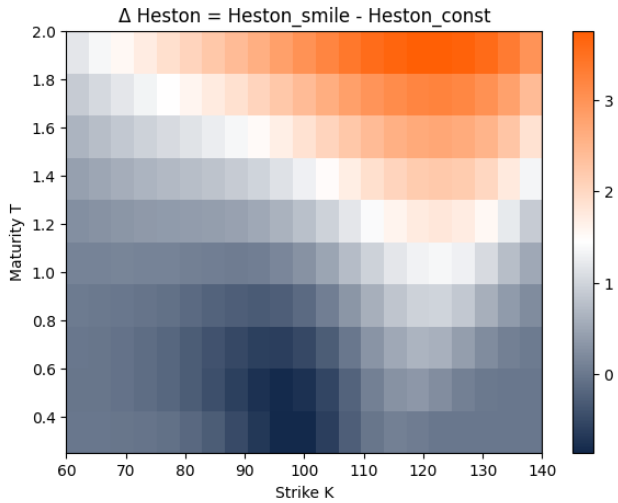

Figure 7: Smile-Based vs Constant-Volatility Heston American Put Prices

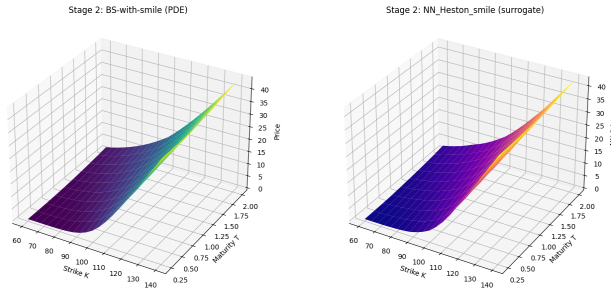

Figure 8: Runtime comparison bar chart

3.6 Neural Network Approximation Results

The neural-network surrogates provide an accurate approximation of the American option pricing surfaces generated in Stage 2. Across both the deterministic local-volatility and Heston settings, predicted prices closely match the corresponding PDE benchmark values over the entire pricing domain.

This strong agreement indicates that the networks successfully learn the global mapping from option parameters to option prices, rather than fitting only local regions of the surface. No systematic bias, such as persistent overpricing or underpricing, is observed, and deviations from the benchmark prices remain limited in magnitude.

3.7 Discussion

Stage 2 demonstrates that volatility structure alone—without introducing stochastic variance—can substantially reshape American option price surfaces. While deterministic local volatility captures smile-induced curvature at low computational cost, it cannot replicate the asymmetric dynamics generated by stochastic volatility models.

At the same time, the neural network results indicate that increased model complexity does not necessarily imply increased approximation difficulty. This finding reinforces the practicality of surrogate pricing approaches for complex option pricing problems.

4 Stage 3: Implied-Volatility Surface from Real Market Data

In this stage, we extend our synthetic experiments to real market data in order to assess whether stochastic- and local-volatility models calibrated from observed option prices can reproduce realistic American option behavior. We follow a strictly structured pipeline, summarized in Section 4.1, and expand each component in detail.

4.1 Pipeline Overview

The Stage 3 workflow follows a structured pipeline that links real-market volatility calibration, PDE-based American option pricing, and neural-network approximation. The main steps are:

1. Acquisition of real market data from NASDAQ AAPL option chains (single-day cross section);
2. Extraction of European Black–Scholes implied volatility from observed American option quotes;
3. Calibration of the Heston stochastic volatility model via European price matching;
4. Evaluation of Heston model fit, calibration error, and structural limitations;
5. Calibration of SVI implied-volatility smiles on a slice-by-slice basis;
6. Construction of an arbitrage-consistent local volatility surface via the Dupire formula;
7. Finite-difference PDE pricing of American put options under:
   - the calibrated Heston stochastic volatility model,
   - the SVI-based local volatility model;
8. Comparison of American option price surfaces across volatility representations;
9. Training and evaluation of a neural-network surrogate using PDE-generated American prices as supervision.

4.2 Real Market Data Source

We use official NASDAQ option chain data for **AAPL American-style equity put options**, collected on a single trading day (Dec 8, 2025). The dataset contains five consecutive monthly maturities:

$$\text{Dec 19th 2025},\ \text{Jan 17th 2026},\ \text{Feb 21th 2026},\ \text{Mar 20th 2026},\ \text{Apr 17th 2026}.$$

Each option chain provides bid, ask, and last prices, trading volume, open interest, and a unified underlying spot quote. Using a single cross-sectional snapshot ensures internal consistency across strikes and maturities and avoids confounding time-series effects.

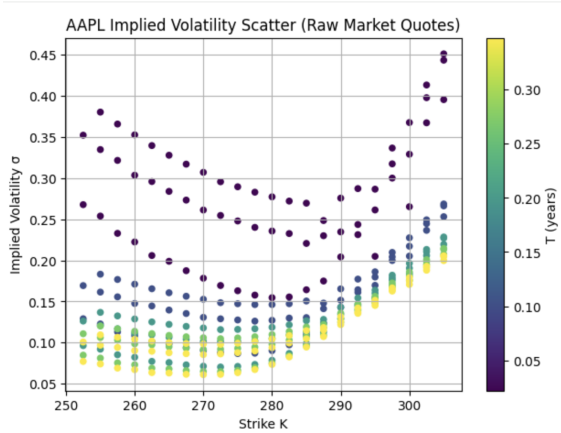

Figure 9: AAPL implied volatility scatter plot from raw market quotes. Colors indicate time to maturity. Short-dated options exhibit strong downside skew, while longer maturities display smoother volatility smiles.

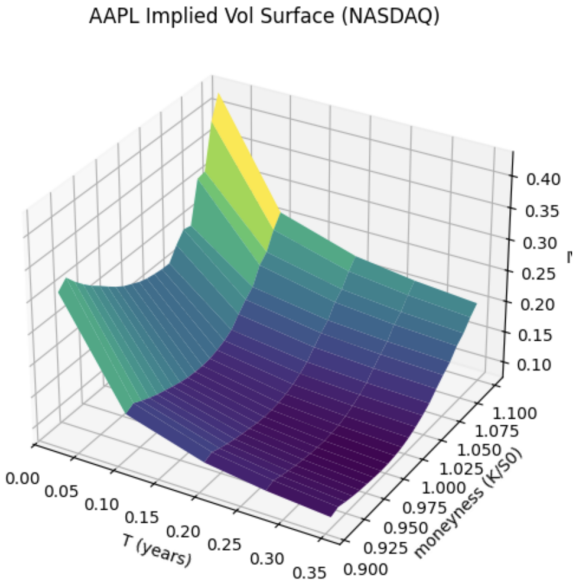

Figure 10: AAPL implied volatility surface constructed from European Black–Scholes implied volatilities. The surface shows pronounced strike-dependent curvature at short maturities and a smoother term structure at longer horizons.

Figure 9 displays the raw implied volatility quotes across strikes and maturities. A pronounced equity skew is evident, with higher implied volatilities for deep out-of-the-money put options. The skew is strongest at short maturities and gradually attenuates as maturity increases, reflecting the dominant role of short-term downside risk in equity option markets.

Figure 10 presents the same information in surface form. The implied volatility surface exhibits strong curvature along the strike dimension at short maturities, together with a clear

term-structure effect as volatility decreases and smooths out for longer maturities. These stylized features highlight the inadequacy of constant-volatility assumptions and motivate the use of more flexible volatility models in subsequent sections.

Taken together, these empirical features indicate that AAPL option prices embed a strong and maturity-dependent volatility structure that cannot be adequately captured by constant-volatility models. To formalize this information in a model-independent manner, we extract the market-implied volatility surface using the European Black–Scholes convention. This implied volatility surface serves as the quantitative benchmark for subsequent model calibration and comparison.

4.3 Heston Calibration via European Price Matching

4.3.1 Calibration Rationale and Methodology

Although the observed AAPL option prices are American-style, closed-form analytic pricing formulas under the Heston stochastic volatility model are available only for European options. A direct calibration of Heston parameters to American prices is therefore not feasible within an analytic framework. To address this mismatch, we adopt a standard industry practice that calibrates the Heston model to European prices implied by the market.

Specifically, the market-implied volatility surface $\sigma_{\mathrm{mkt}}(K,T)$ is first converted into European Black–Scholes prices,

$$C^{\mathrm{BS}}(K,T;\sigma_{\mathrm{mkt}}),$$

which provide a model-independent representation of market prices under the European payoff assumption. For a candidate Heston parameter vector

$$\theta_H = (\kappa, \theta, \sigma_v, \rho, v_0),$$

the corresponding Heston European option prices

$$C^H(K,T;\theta_H)$$

are computed using the closed-form Heston pricing formula.

The Heston parameters are obtained by solving the following least-squares optimization problem:

$$\theta_H^* = \arg\min_{\theta_H} \sum_{i=1}^{N} \left(C_i^H(\theta_H) - C_i^{\mathrm{BS}} \right)^2,$$

where the summation is taken over all selected strikes and maturities in the calibration set.

To reduce noise arising from extremely deep out-of-the-money contracts and illiquid quotes, the calibration domain is restricted to the moneyness range

$$0.85 \leq K/S_0 \leq 1.20.$$

4.3.2 Calibration Results and Parameter Interpretation

After implementing the European price-matching calibration procedure described above, we obtain a single set of Heston parameters that best fit the market-implied European price surface in a least-squares sense. Table 10 reports the calibrated parameter values obtained from NASDAQ AAPL option data.

Table 10: Calibrated Heston Parameters for AAPL

Parameter	Value
Mean reversion speed κ	7.2334
Long-run variance θ	0.0301
Volatility of volatility σ_v	4.8860
Correlation ρ	0.7452
Initial variance v_0	0.0476

The calibrated parameters are economically interpretable. The large volatility-of-volatility σ_v indicates substantial short-term variance fluctuations, while the positive correlation ρ reflects the pronounced downside skew commonly observed in single-stock equity options. The estimated mean-reversion speed κ suggests moderately fast reversion of variance toward its long-run level.

To quantify the overall calibration quality, Table 11 reports the root mean squared error (RMSE) between Heston European prices and Black–Scholes European prices implied by the market.

Table 11: Heston Calibration Error (European Prices)

Metric	Value
RMSE	0.678

The calibrated Heston parameters are subsequently used in a finite-difference PDE framework to price American options.

4.3.3 Heston Model Fit Evaluation and Limitations

The Heston model is able to reproduce the broad convexity and overall skew direction of the AAPL implied-volatility surface. However, the remaining pricing discrepancy is non-trivial, with an overall calibration error of

$$\text{RMSE} \approx 0.67.$$

This error reflects intrinsic structural limitations of the model rather than numerical instability or implementation issues.

In particular, AAPL volatility exhibits pronounced jump risk and earnings-related spikes that are not captured by the continuous diffusion dynamics assumed in the Heston framework. Moreover, the observed implied-volatility surface displays strong local convexity at short maturities, which cannot be fully accommodated by a single five-parameter stochastic volatility model. These challenges are further amplified by market microstructure noise and the typically stronger skew observed in single-stock options relative to index options.

To verify that the residual error is structural in nature, we perform several sanity checks. The calibrated Heston parameters are stable across different initializations and remain within economically plausible ranges. In addition, the implied-volatility smiles generated by the model preserve the correct skew direction and term-structure ordering across maturities, confirming that the calibration procedure is well-behaved.

Overall, while the Heston model provides a reasonable first-order approximation to the AAPL implied-volatility surface, it cannot fully reproduce all observed market features. This motivates the use of more flexible, arbitrage-free volatility representations, such as SVI-based smile models and local volatility surfaces, which we introduce next.

4.4 SVI-Based Local Volatility Construction

4.4.1 SVI Smile Calibration

Motivated by the limited flexibility of parametric stochastic volatility models, we adopt the Stochastic Volatility Inspired (SVI) parameterization to obtain a more expressive representation of the implied-volatility surface. The SVI model is specifically designed to capture strong skew and local curvature commonly observed in single-stock equity options.

Following standard practice, we work with total implied variance

$$w(k,T) = \sigma^2(K,T)\,T, \qquad k = \log(K/S_0),$$

and calibrate the raw SVI functional form

$$w(k) = a + b\left(\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\right)$$

using nonlinear least squares. Calibration is performed independently for each maturity slice, allowing the implied-volatility smile to adapt flexibly across maturities without imposing global parametric restrictions.

The calibrated SVI representation provides a smooth and arbitrage-consistent interpolation of the market implied-volatility surface, which serves as the input for local volatility construction.

4.4.2 Local Volatility Surface Construction

Using the calibrated SVI implied-volatility surface, we construct the local volatility surface via the Dupire formula. Given total implied variance $w(k,T)$, the local volatility is defined as

$$\sigma^2_{\text{loc}}(k,T) = \frac{\partial_T w(k,T)}{\left(1 - \frac{k}{w}\partial_k w\right)^2 + w\,\partial_{kk} w},$$

where derivatives with respect to log-moneyness are obtained analytically from the SVI functional form, and the time derivative $\partial_T w$ is approximated numerically using finite differences across adjacent maturity slices.

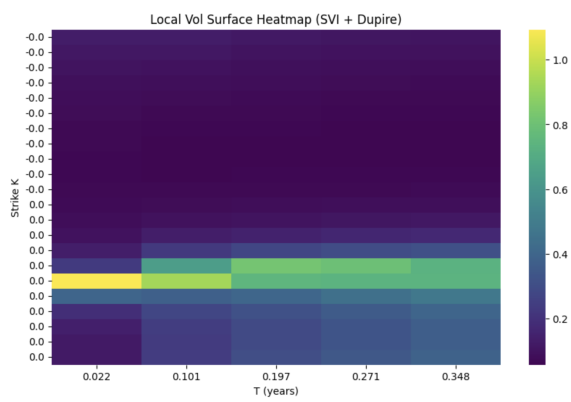

Figure 11: Local volatility heatmap constructed via SVI calibration and the Dupire formula. Elevated local volatility concentrates near at-the-money for short maturities, consistent with strong short-dated smile curvature.

Figure 11 highlights pronounced local volatility peaks near at-the-money at short maturities. This pattern reflects the sharp local curvature observed in short-dated implied-volatility smiles and indicates that instantaneous volatility must increase locally to reproduce the observed market surface.

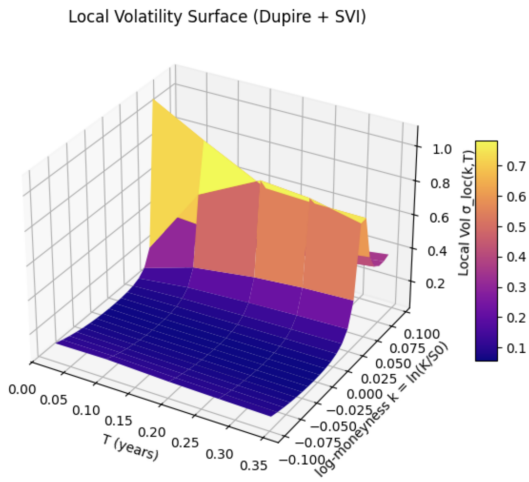

Figure 12: Three-dimensional view of the local volatility surface obtained from SVI-based implied volatilities using the Dupire framework. The surface becomes smoother as maturity increases, indicating more stable long-term variance dynamics.

Figure 12 provides a three-dimensional view of the local volatility surface. The most pronounced structure appears at short maturities, while longer maturities exhibit smoother dependence on log-moneyness, consistent with the term-structure smoothing observed in the market implied-volatility surface.

Overall, the Dupire local volatility surface is, by construction, consistent with the observed implied-volatility surface while remaining sufficiently smooth for finite-difference PDE pricing of American options, which we consider in the following section.

4.5 PDE Pricing of American Options

To assess how different volatility representations translate into early-exercise values, we price AAPL American put options under two finite-difference PDE frameworks:

- **Heston stochastic volatility PDE**, using the calibrated Heston parameters from Section 4.3;
- **SVI-based local volatility PDE**, where the local volatility surface is constructed via SVI calibration and the Dupire mapping in Section 4.4.2.

For comparability, both solvers are run on the *same strike–maturity grid* extracted from the option chain, with identical finite-difference resolution settings (spatial and time discretization). The only difference is the model dynamics (stochastic volatility vs. deterministic local volatility).

Implementation summary. For each grid point (K, T), we solve the corresponding American put pricing problem using QuantLib finite-difference engines. Under Heston, the pricing engine solves the two-dimensional Heston PDE with early-exercise constraints. Under local volatility, we solve the Black–Scholes PDE with a volatility term structure obtained from the SVI-implied surface (consistent with the constructed local volatility representation). This yields two American price surfaces:

$$V^{\mathrm{Am}}_{\mathrm{Heston}}(K, T), \qquad V^{\mathrm{Am}}_{\mathrm{LocVol}}(K, T).$$

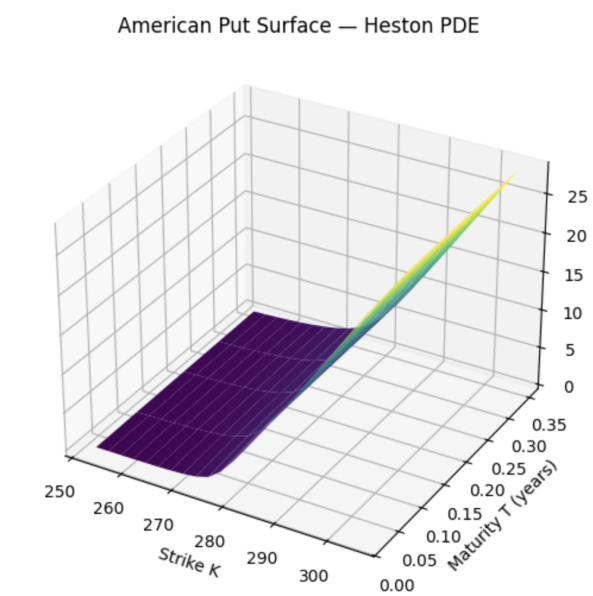

Figure 13: American put price surface obtained from the Heston PDE solver on the market strike–maturity grid.

Figure 13 shows the American put surface under the calibrated Heston dynamics. Prices increase with strike, reflecting higher intrinsic value for deeper in-the-money puts, and increase with maturity due to larger time value and a greater likelihood of beneficial early exercise.

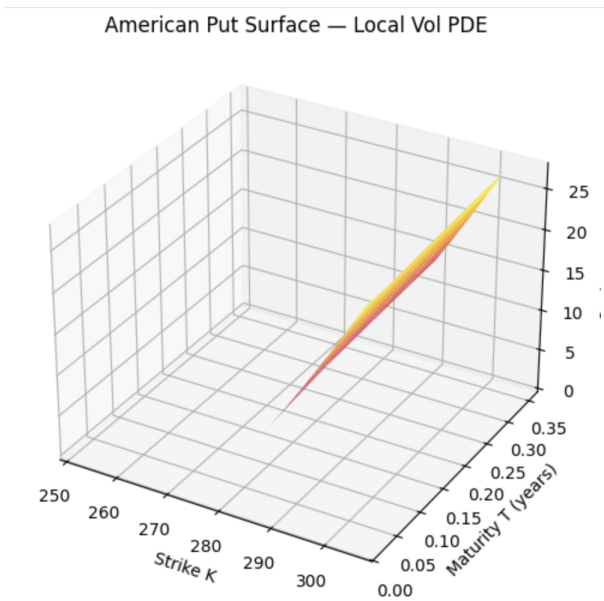

Figure 14: American put price surface obtained from the local volatility (SVI + Dupire) PDE solver on the same strike–maturity grid.

Figure 14 reports the corresponding American put surface under the SVI-based local volatility framework. Because the local volatility surface is constructed to be consistent with the observed implied-volatility surface, this PDE solution reflects stronger strike- and maturity-local curvature inherited from the market smile structure.

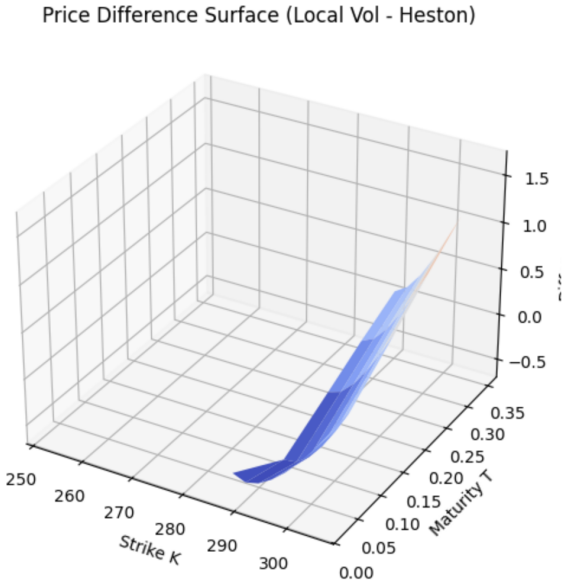

Figure 15: Difference surface $V^{\mathrm{Am}}_{\mathrm{LocVol}}(K,T) - V^{\mathrm{Am}}_{\mathrm{Heston}}(K,T)$ on the market strike–maturity grid.

Figure 15 visualizes the model discrepancy by plotting the pointwise difference between the two American price surfaces. The difference is small in regions where both models imply similar effective volatility, and becomes more pronounced where the local volatility surface exhibits sharper short-dated curvature or wing behavior.

Taken together, the PDE-based results indicate that differences in volatility modeling translate systematically into differences in American option values once early-exercise features are incorporated. While both PDE frameworks produce smooth and economically reasonable price surfaces, the local volatility PDE responds more strongly to short-dated smile curvature and strike-local features inherited from the implied-volatility surface. In contrast, the Heston PDE produces a smoother price surface driven by a small set of global stochastic-volatility parameters, which limits its ability to match localized market structure.

These observations suggest that, even when calibrated to the same market data, the choice of volatility representation plays a central role in determining American option prices under PDE methods. The resulting PDE price surfaces therefore provide a natural and informative benchmark for evaluating fast surrogate models.

In the next section, we use these PDE-generated American prices as supervised learning targets for a neural-network surrogate model and evaluate predictive accuracy using MAE and RMSE.

4.6 Neural Network Surrogate Training

Having constructed American option prices under both the Heston PDE and the SVI-based local volatility PDE (calibrated from real AAPL market data), we next train a neural-network surrogate to approximate the resulting pricing surfaces. Unlike Stage 2 synthetic experiments, the targets here inherit *real-market imperfections*—microstructure noise, sparse OTM liquidity, and calibration mismatch—so validation errors in dollar terms are expected to be higher and more spatially structured.

Training setup. We train a supervised surrogate using PDE-generated American put prices as labels. Inputs are moneyness K/S_0, maturity T, and a model indicator ($0 =$ Heston PDE, $1 =$ SVI-local-vol PDE). The output is the American option price. We restrict the domain to

$$0.8 \leq K/S_0 \leq 1.2$$

to reduce the impact of extremely deep OTM quotes and unstable extrapolation, and we remove non-finite targets (NaN/Inf) prior to training. Both inputs and targets are standardized. The surrogate is a feed-forward MLP with two hidden layers (64, 64) and ReLU

activations, trained with Adam and early stopping.

Overall predictive accuracy. Table 12 reports validation performance in price space. The surrogate achieves sub-dollar RMSE overall. Note that these errors reflect not only approximation error of the network, but also the fact that PDE-generated prices are based on calibration to *real* implied-volatility data (American market quotes converted into a European representation, then calibrated), so residual model mismatch propagates into the training labels.

Table 12: Neural Network Surrogate Performance (Validation Set)

Training Source	Sample Size	MAE	RMSE
Heston PDE	25	0.334	0.447
SVI Local Vol PDE	7	0.354	0.466
Overall	32	0.339	0.451

Slice-level validation. To qualitatively assess shape fidelity, we compare NN predictions against PDE prices on a representative maturity slice ($T \approx 0.197$). Figure 16 shows the Heston case, while Figure 17 shows the SVI-local-vol case. In both settings, the surrogate tracks the monotone American-put curve well, with slightly larger deviations around regions where the early-exercise boundary changes most rapidly across strikes.

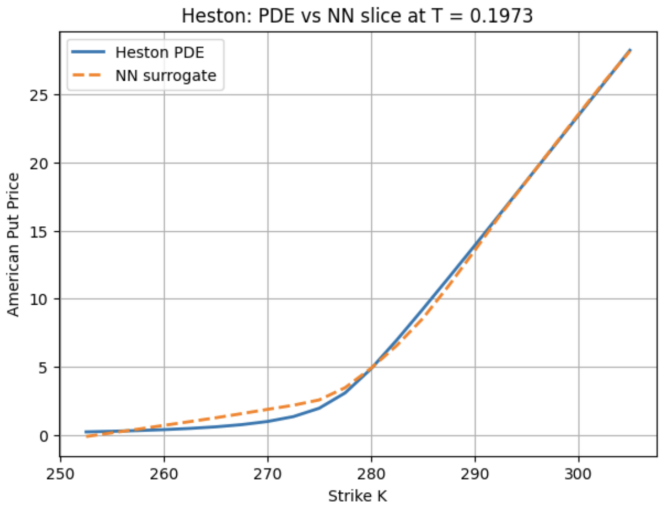

Figure 16: Heston case: PDE price curve vs. neural-network surrogate on a representative maturity slice ($T \approx 0.197$).

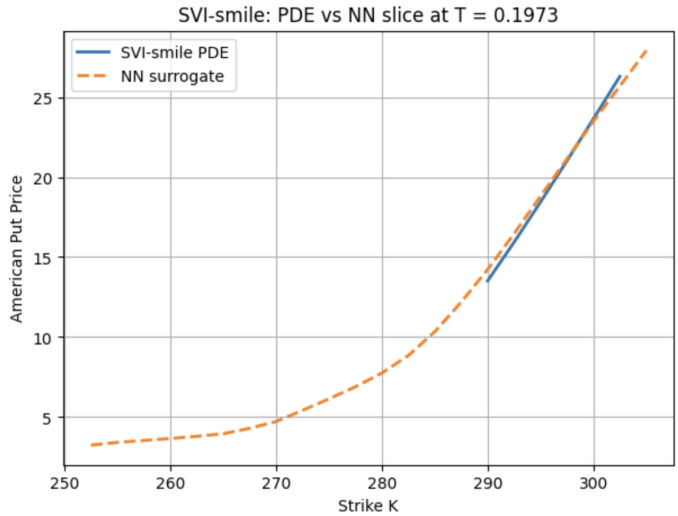

Figure 17: SVI-local-vol case: PDE price curve vs. neural-network surrogate on the same maturity slice ($T \approx 0.197$).

Model difference and spatial error structure. Figure 18 plots the pointwise difference between the two PDE-generated price curves at the same maturity. The difference changes sign across strikes, indicating that the SVI-based local volatility PDE introduces localized price adjustments relative to the smoother parametric Heston PDE, consistent with stronger short-dated smile curvature.

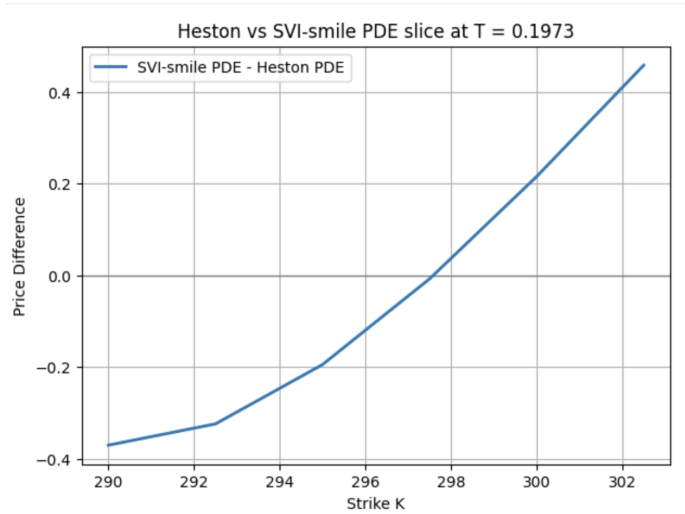

Figure 18: Price difference on the representative maturity slice: SVI-local-vol PDE minus Heston PDE ($T \approx 0.197$).

We further visualize NN errors over the full (K, T) grid via heatmaps in Figure 19–20. The color shows NN − PDE (price space). Errors are generally smooth and bounded, while slightly larger deviations appear in short maturities and near boundary regions where American exercise decisions are most sensitive.

Importantly, the *blank (white) region* in the SVI-local-vol heatmap corresponds to missing or invalid PDE labels rather than a plotting artifact. In our pipeline, some SVI-local-vol PDE prices become non-finite (NaN) for certain (K, T) points due to illiquid contracts and/or numerical instability in the smile-to-surface construction (e.g., sparse strike coverage at particular maturities). These points are filtered out before training; when the error map is plotted back on the original grid, those missing entries naturally appear as blank cells.

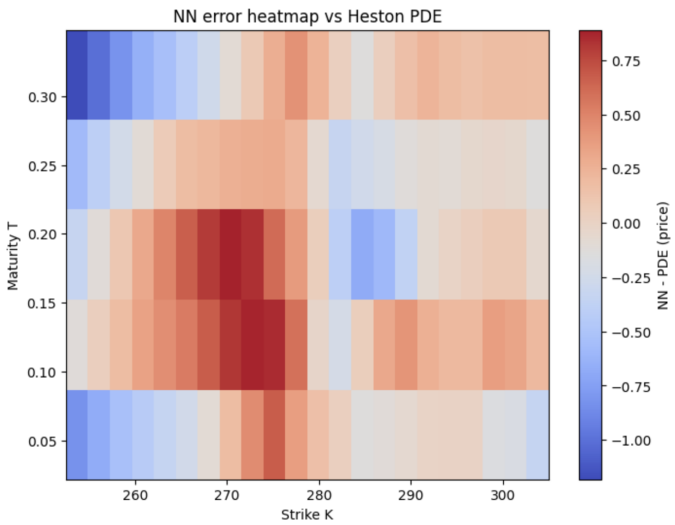

Figure 19: Neural-network error heatmap relative to Heston PDE targets (NN − PDE).

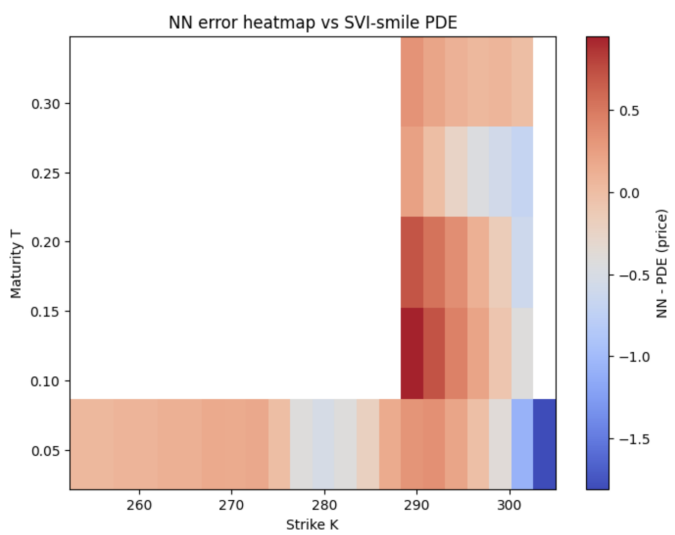

Figure 20: Neural-network error heatmap relative to SVI-local-vol PDE targets (NN−PDE). Blank regions correspond to missing/invalid PDE labels removed prior to training.

Overall, the results demonstrate that the neural network provides an accurate and stable surrogate for American option prices generated by both PDE frameworks. Across the relevant

moneyness–maturity region, the surrogate preserves the shape, monotonicity, and relative curvature of the underlying PDE price surfaces.

The remaining prediction errors are not uniformly distributed but concentrate in regions where American option values are most sensitive, namely near the early-exercise boundary and short maturities. In addition, because the training labels are derived from PDE solvers calibrated to real market implied-volatility data, residual noise and model misspecification are inevitably propagated into the learning targets. These effects explain the observed error levels and spatial patterns, rather than limitations of the neural-network architecture itself.

Taken together, the results confirm that the proposed surrogate is sufficiently accurate for fast approximation of American option prices under both Heston and SVI-based local volatility models, while faithfully reflecting the structural differences between the two PDE frameworks.

4.7 Stage 3 Summary

Stage 3 extends the American option framework from synthetic experiments to real market data and highlights the interplay between volatility modeling, PDE-based pricing, and neural-network approximation.

We first show that, although the Heston stochastic volatility model provides a reasonable first-order description of market behavior, it is unable to fully reproduce the strong local curvature and short-dated skew observed in single-stock implied volatility surfaces. Calibration errors persist even under stable and economically plausible parameters, reflecting structural model limitations rather than numerical deficiencies.

By contrast, the SVI-based local volatility framework offers a substantially more flexible and market-consistent representation of implied volatility. Through slice-by-slice SVI calibration and Dupire inversion, the resulting local volatility surface accurately reflects short-maturity curvature and wing behavior, leading to systematic differences in American option prices relative to the Heston PDE. These differences are localized in regions where volatility smiles and early-exercise effects interact most strongly.

Using PDE-generated American prices from both models as training targets, we demonstrate that a neural network surrogate can accurately learn the underlying pricing surfaces across the relevant moneyness–maturity domain. While prediction errors are non-negligible in absolute terms, they are concentrated near early-exercise boundaries and regions of heightened market uncertainty, and are consistent with label noise and model mismatch inherent in real-market implied-volatility calibration. Importantly, the surrogate preserves the structural distinctions between Heston and SVI-local-vol pricing rather than smoothing them

away.

Overall, Stage 3 establishes a coherent pipeline from real-market volatility calibration to American option PDE pricing and fast neural-network surrogates. The results illustrate both the limitations of parametric stochastic volatility models and the advantages of more flexible smile-based approaches, while demonstrating the practicality of neural surrogates for efficient American option valuation in realistic market settings.

5 Discussion

To better understand how volatility structure influences the learnability of American option pricing, we compare the neural network's performance across three stages. Because the Heston model is the only model present in all three settings, it provides a consistent basis for cross-stage comparison. All comparisons that follow are made within the Heston model framework.

A natural starting point is to evaluate the neural network in the familiar price space and compare MAE and RMSE from PDE-based valuations.

Table 13: Type A: Test-set Heston NN Performance (Price Space)

Model	**MAE**	**RMSE**
Stage 1 (constant vol)	0.068792	0.088819
Stage 2 (synthetic smile)	0.069357	0.091384
Stage 3 (market calibrated)	0.175947	0.217581

In price space, Stage 1 and 2 produce nearly errors, with MAE around 0.069. This indicates that the network easily captures smooth and monotonic synthetic pricing surfaces. In contrast, Stage 3 exhibits higher MAE and RMSE (0.176 and 0.218), reflecting the greater complexity of market-generated surfaces. This confirms the expectation that incorporating real-world implied-volatility curvature increases the difficulty of pointwise approximation.

While price-level metrics describe local differences, they do not capture how well the neural network reproduces the overall shape of the pricing surface. Therefore, next we start to examine global surface-level errors and distributional patterns across strike–maturity grids.

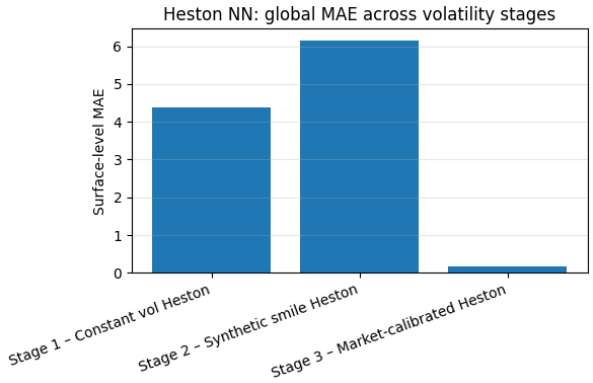

Figure 21: Global MAE Across Stages

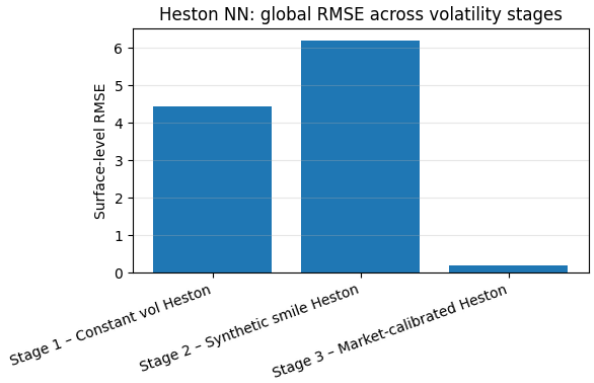

Figure 22: Global RMSE Across Stages

When measuring global deviations between the NN surface and PDE surface, the constant-volatility and synthetic-smile regimes produce large, stable errors (4–6 range), while the market-calibrated surface produces much smaller deviations (mostly below 0.4). This inverted relationship, higher pointwise MAE but much smaller global surface error, highlights that the structure of the pricing surface matters more than the raw volatility level. This implies NN learns better when the surface has clear shape. We guess the flat surfaces don't give the model enough signal to figure out what to do.

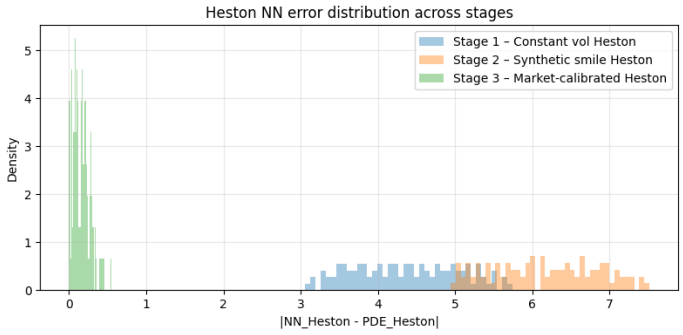

Figure 23: Error Distribution Across Stages

The error distribution plot further confirms this pattern: Stage 1 and Stage 2 produce large, tightly clustered errors in the 3–7 range. This reflects a systematic surface-level shift, while Stage 3 errors remain an order of magnitude smaller and much more concentrated near zero. This separation of distributions visually reinforces the structural advantage of the market-calibrated surface.

The surface-level results raise an important question: why do certain volatility regimes produce large, smooth errors while others produce smaller, more localized deviations? To answer this, we analyze the curvature of the pricing surface.

Table 14: Gamma Buckets and Error Statistics Across Stages

Stage / Bucket	**Mean γ**	**MAE Error**	**95% Error**	**Count**
Stage 1 – Constant vol Heston (flat surface)	0.000000	4.365442	5.300972	60
Stage 2 – Synthetic smile Heston (flat surface)	0.000000	6.169086	7.084605	60
Stage 3 – Market-calibrated Heston (Q0–20%)	0.001100	0.174747	0.309778	12
Stage 3 – Market-calibrated Heston (Q20–40%)	0.002312	0.116615	0.202694	12
Stage 3 – Market-calibrated Heston (Q40–60%)	0.005024	0.146988	0.273815	12
Stage 3 – Market-calibrated Heston (Q60–80%)	0.015283	0.245383	0.436525	12
Stage 3 – Market-calibrated Heston (Q80–100%)	0.070918	0.161124	0.285207	12

The curvature–error analysis explains this phenomenon. In Stage 1 and Stage 2, the pricing surfaces are nearly flat in strike, producing curvature close to zero. The NN struggles in such low-signal regimes because the mapping lacks informative gradients; many function values appear interchangeable, leading to large but smooth misspecification. By contrast, the market-calibrated surface exhibits meaningful curvature across strikes, providing the network with richer local structure to learn from. These findings suggest that surface geometry, not volatility magnitude, determines learnability.

Taken together, these three layers of analysis provide a coherent explanation of the network's strengths and limitations across volatility regimes.

6 Conclusion

This project examined how well a neural network can approximate American put prices under three progressively richer volatility regimes:

- a constant-volatility Heston model,
- a synthetic smile surface, and
- a Heston surface calibrated to real market data.

Across all stages, we benchmarked the neural network against high-quality PDE-based pricing and evaluated performance in both price space and surface space. Taken together, the results show that volatility structure plays a decisive role in shaping the learnability of the pricing function.

Overall, our results indicate that neural networks are reliable surrogate pricers when the underlying pricing surface exhibits sufficient structure, as in the market-calibrated Heston case. Flat or weakly informative surfaces, on the other hand, limit the model's ability to anchor predictions, even if the volatility model itself is simpler.

Future work may explore:

- architectural inductive biases that explicitly capture curvature, or
- hybrid approaches that combine deep networks with local PDE features

enabling more robust performance across heterogeneous volatility regimes.

7 References

Example citation entries:

- QuantLib Python Cookbook, Section 22.
- F. Black and M. Scholes (1973): "The Pricing of Options and Corporate Liabilities."
- J. Hull: *Options, Futures, and Other Derivatives.*
- Longstaff & Schwartz (2001): LSM Monte Carlo.

8 Appendix

8.1 Colab Code Repository

The full implementation and experimental notebooks are available on Google Colab at:

`https://colab.research.google.com/drive/1L6hf_1QlxrQb1t11-WLvL5CFFC0vRYHR`